# MS2 Report

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## Vision

Our vision remains largely unchanged since MS1 because we have made considerable progress that points to future success; we are comfortable with the goals we have for now. Our current version of this system is one that trains a Q-Learning RL agent to make buy, sell, hold decisions on historical stock market data. It learns profitable trading strategies through trial and error. The user interacts with an interface where they can select stock datasets and configure training parameters. Once it is trained, we can test the agent on market data, with its performance visualized through a chart showing portfolio value and trade history over time. This system will include baseline strategies, like buy-and-hold, random, and simple moving average, so that the RL agent's performance can be directly compared with conventional approaches.

## Progress Summary

We worked on creating all the baseline strategies; this includes buy and hold, random policy, and simple moving average. We also created all the .mli files, so we have set up the functionality we want for our next sprints. We made test cases for our baseline strategies and gathered data to use for these tests using a Python script and yFinance (Python package to get information from Yahoo Finance). Finally, we have everything printed out neatly in a terminal; it asks for user input for the data to be used and then prints the results of the baseline strategies.

## Activity Breakdown

- Anmol

    - Work Completed: Test cases and environment.ml
    - Hours Spent: 4
- Pratyush
    - Work Completed: Baseline strategies
    - Hours Spent: 3
- Rishabh
    - Work Completed: Wrote script to gather data and agent.ml
    - Hours Spent: 4
- Simran
    - Work Completed: Metrics.ml
    - Hours Spent: 3

**Productivity Analysis**

We were productive as a team; we accomplished almost everything we wanted to. The only thing we didn't do was create a graph, but that was more a functional choice on our end. We opted for a terminal interface for this stage, as there isn't much to show through a graph right now. We estimated how much work this stage of the project would be quite well. If we can replicate this for our next sprint, we will be in good shape. In the next iteration we will focus our productivity on wrapping up the model and also creating a helpful UI once we are confident in our backend functionality.